

REVIEW ARTICLE

A Century of Elk Restoration in Eastern North America

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Abstract

Over a century has passed since elk were extirpated in eastern North America. During that time, numerous attempts to reintroduce elk into eastern North America have resulted in varying degrees of success and failure. An overview of restoration efforts during the last 100 years is presented here with emphasis on the differences in rates of population change among regions and differences in major causes of elk mortality during both the pre- and post-acclimation

periods. Approximately 40% of recorded elk reintroduction attempts in eastern North America resulted in failure, with the majority of these having occurred in the first half of the 20th century. Although rates of population change in elk were highly variable, they were not related to founding population size. Major causes of mortality varied among regions and should be considered in future reintroduction attempts.

Key words: *Cervus elaphus*, management, mortality, population growth rates, reintroduction, restoration success.

Introduction

Prior to the arrival of the Europeans, it was estimated that 10 million elk were present in North America (Seton 1927). Historically the combined ranges had six subspecies (Roosevelt *Cervus elaphus roosevelti*, Tule *C. e. nannodes*, Rocky Mountain *C. e. nelsoni*, Merriam's *C. e. merriami*, Manitoban *C. e. manitobensis*, and Eastern *C. e. canadensis*) that occupied the majority of North America (O'Gara & Dundas 2002) (Fig. 1). The eastern elk was extirpated by 1867 and Merriam's elk became extinct by the early 1900s (O'Gara 2002). The reduction in elk populations and range has been primarily attributed to overexploitation and habitat loss (O'Gara & Dundas 2002). By the late 1970s, approximately 500,000 elk were primarily found in the western part of the continent (Bryant & Maser 1982). Today, it is estimated that there are slightly more than 1 million elk across North America in several scattered populations (Fig. 1).

Reintroducing animals is a common conservation and management tool that has proven to work well for many species, especially native game species (Griffith et al. 1989), and reintroductions have occurred for over 100 years (Kleiman 1989). Since the extirpation of elk from the east, numerous reintroductions have been attempted with varying degrees of success

and failure (O'Gara & Dundas 2002). Although many elk reintroductions have taken place since the early 1900s, sufficient monitoring has been lacking and related publications are sparse (Larkin et al. 2001). For example, the Pennsylvania elk herd was not studied until 60 years after its initial release (Eveland et al. 1979).

The objective of this article is to provide an overview of elk restoration attempts in eastern North America by reviewing the current literature and survey information obtained from regional elk managers. The goals are to examine major causes of mortality and to detail the trends in population growth by comparing rates of increase among regions.

Methods

In order to provide an overview of the population trends of elk reintroductions in eastern North America, survey questionnaires were e-mailed to primary elk managers in each respective region during September 2013. Pertinent literature and management reports were gathered as supplementary material. Eastern North America was defined as any province or state east of the Great Plains. On the basis of the information gathered from surveys, reports, and the literature, the exponential rates of increase ($r = \ln(N_{t+1}/N_t)$) were calculated using regression analysis, which incorporated available population estimates since the time of reintroduction up to the most recent population estimate. A linear regression was used to examine the relationship between the number of elk released and the exponential rate of increase. Martin (2011) found that reintroduced elk in Ontario took 1–3 years to acclimate, or habituate to their environments based on spatial behaviors. Causes of mortality during the release years and/or within the first 3 years post-release

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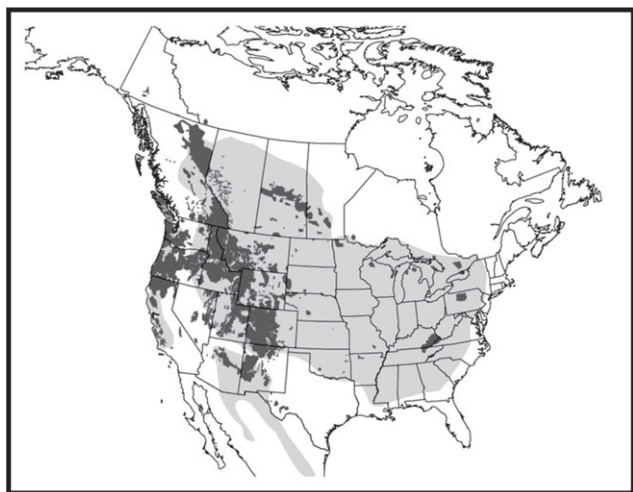


Figure 1. Historical (light gray) and current (dark gray) elk range in North America.

(acclimation phase) were thus calculated for each elk population with available information. More recent major sources of post-acclimation mortality were also detailed for each elk population.

Reintroductions: An Overview

Approximately 40% of documented elk reintroductions in eastern North America have resulted in failure within 5–94 years (Table 1). Elk population declines have been attributed to a variety of factors including vehicle collisions, poor release conditions, lack of appropriate habitat, hunting or poaching, crop damage, disease and parasites, and less commonly poor management (Witmer 1990; O’Gara & Dundas 2002). More recent attempts have resulted in varying degrees of success. The rate of population change over time (r) for established reintroduced elk populations in eastern North America ranged from -0.05 to 0.13 (Table 1), with the populations from Kentucky, United States and Bancroft, Ontario increasing at the most rapid rates. The majority of populations have grown since reintroduction; however, the populations from Minnesota, United States and Lake of the Woods (LOW), Ontario, have decreased in size (Fig. 2; Table 1). Hunting seasons have been opened in several regions and contribute to population control and potentially account for slower rates of increase (i.e. Minnesota, United States); however, other populations have low growth rates or are declining without the presence of a legal hunt (i.e. Burwash, LOW, Ontario and Minnesota).

Table 1. Statistics on elk released in different regions of eastern North America.

<i>Region</i>	<i>Total Released</i>	<i>Year(s) Released</i>	<i>2013 Estimate</i>	<i>Years Since Release</i>	<i>r</i>	<i>First Hunt</i>
Kentucky	1,547	1995–2002	10,000	11	0.13	2001
Bancroft, ON	120	1998–2001	500	12	0.13	2011
LH-NS, ON	47	1998–2001	200	12	0.12	NA
Wisconsin	25	1995	154	18	0.12	NA
North Carolina	52	2001–2002	150	11	0.10	NA
Tennessee	201	2000–2003	400 ^a	10	0.08	2009
Arkansas	112	1981–1985	617	28	0.06	1998
Michigan	23	1914–1915	1050	97	0.04	1920s
Pennsylvania	177	1913–1926	833	87	0.04	2001
Burwash, ON	172	1998–2001	145	12	0.03	NA
Minnesota ^b	56	1914	28	98	-0.01	1987
LOW, ON	104	1998–2001	60	12	-0.05	NA
Missouri	108	2011–2013	110	NA	NA	NA
Virginia	18	2012	24 ^a	NA	NA	NA
Failed attempts						
<i>Region</i>	<i>Total Released</i>	<i>Year(s) Released</i>	<i>2013 Estimate</i>	<i>Years Since Release</i>	<i>Extirpation Date</i>	
Alabama	55	1916	NA	5	1921	
Arkansas	11	1933	NA	~20	1950s	
Florida	6	1968	NA	5	1973	
Indiana	UNK	Early 1960s	NA	UNK	UNK	
Louisiana	20	1916	NA	UNK	UNK	
Missouri	10	1951	NA	8	1959	
New Hampshire	12	1903	NA	94	1997	
New York	332	1893–1906	NA	60	1953	
Virginia	110–150	1917	NA	~50	1960s	
Ontario	UNK	Early/mid 1900s	NA	UNK	Trace remained	

LH-NS, Lake Huron-North Shore; UNK, unknown; r , exponential rate of increase; NA, not applicable.

Failed attempts; source: O’Gara and Dundas (2002).

^a2012 estimate.

^bGrygla herd.

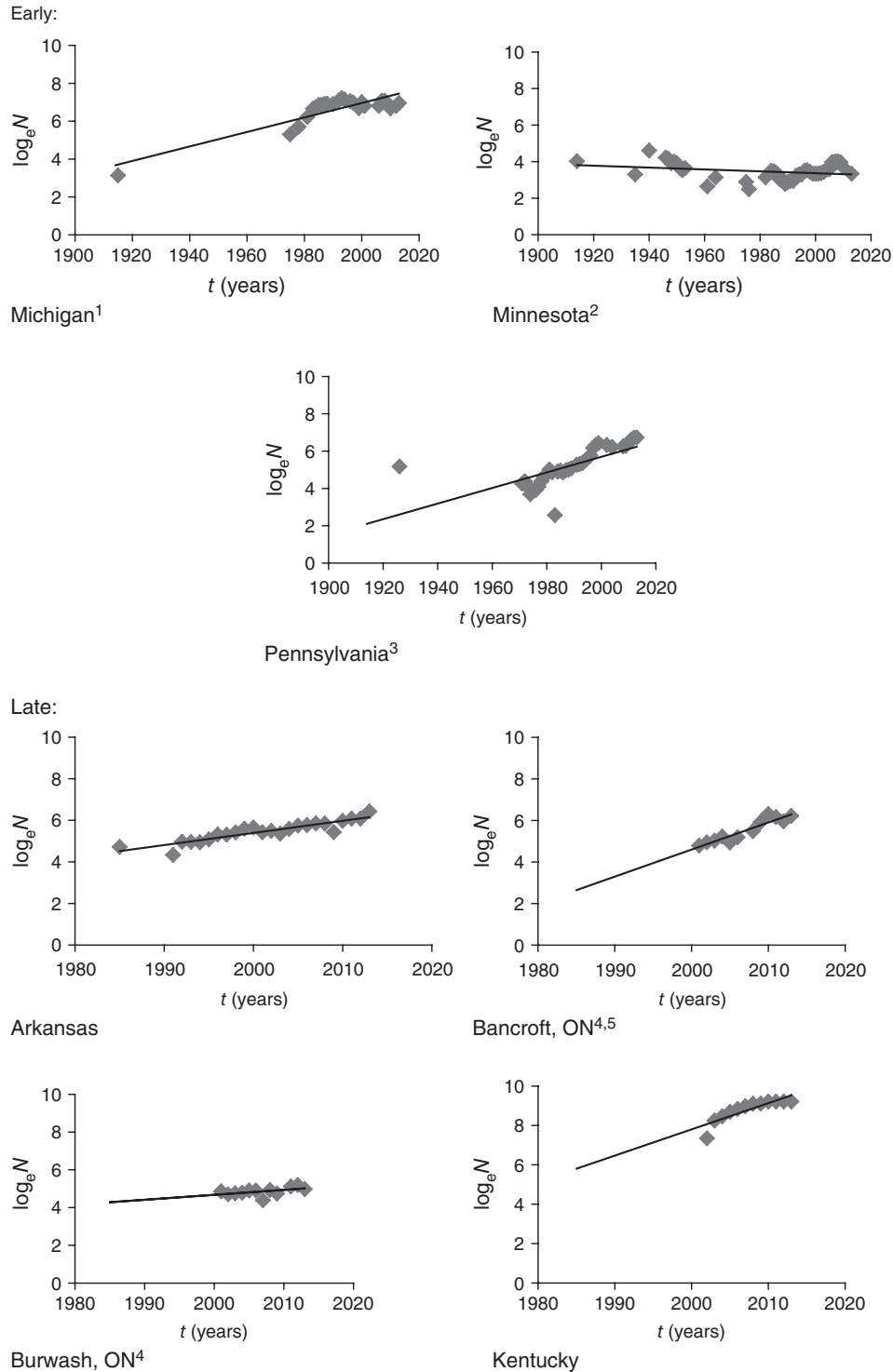


Figure 2. Estimates of elk population growth in different restoration regions across eastern North America. ^aLake Huron-North Shore, ^bLake of the Woods. ¹Michigan Department of Natural Resources. 2012. Michigan Elk Management Plan. Lansing, Michigan. ²Minnesota Department of Natural Resources. 2009. Strategic Management Plan for Elk Minnesota Department of Natural Resources November 2, 2009, Minnesota. ³Banfield, J., Perlock, E., and C. Rosenberry. 2013. Elk Research/Management. Pennsylvania Game Commission Bureau of Wildlife Management Project Annual Job Report; DeBerti, J.M. 2006. Management Plan for Elk in Pennsylvania 2006–2016. Northcentral Regional Office Pennsylvania Game Commission. ⁴Ontario Ministry of Natural Resources Annual Provincial Elk Status Updates. ⁵Rosatte, R. *In Press*. The Behaviour and Dynamics of a Restored Elk (*Cervus canadensis manitobensis*) Population in Southern Ontario, Canada: 5–12 Years Post Restoration. Canadian Wildlife Biology and Management. ⁶Stowell, L.R., Zickmeister, M., Jonas, K.W. Wallengang, K., Roepke, S.C., Gilbert, J., Eklund, D.A., Ginnett, T. Rolley, R., Wydeven, A., Dhuey, B., Babros, T., and K. Johansen. 2012. 2012 Clam Lake and Black River Elk Management Plan Amendment. Wisconsin Department of Natural Resources. Madison, Wisconsin.

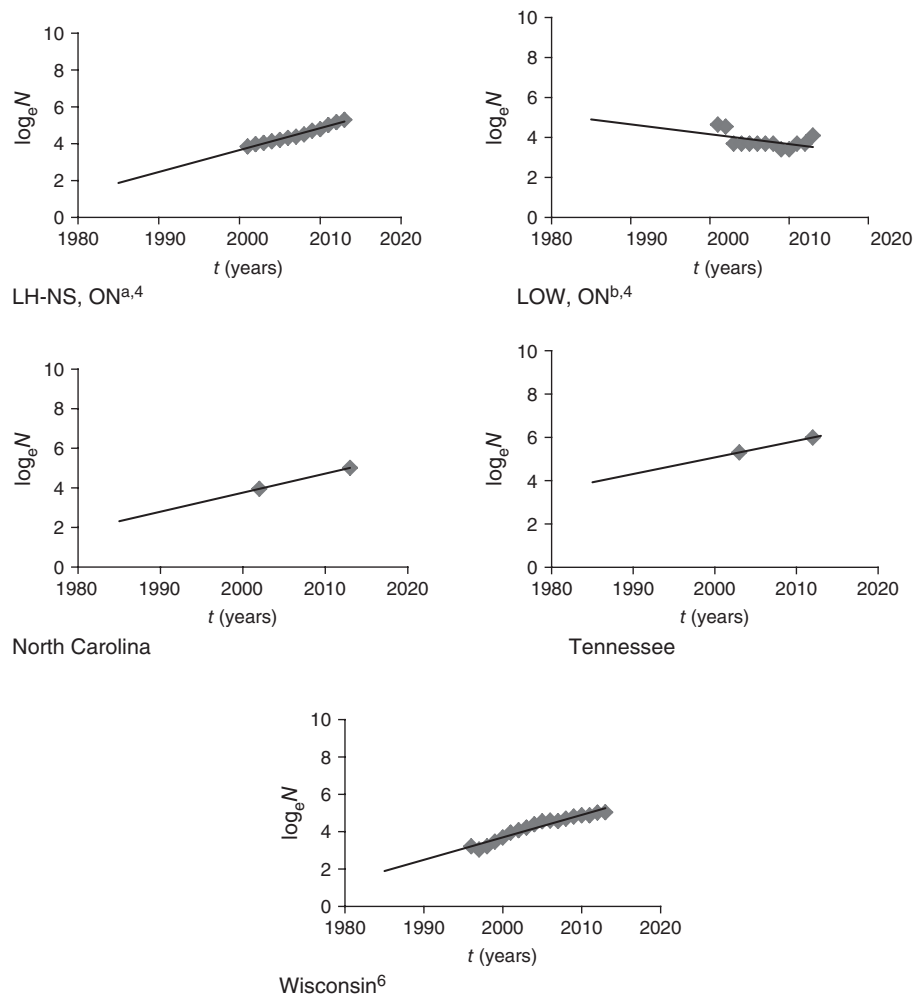


Figure 2. Continued.

Griffith et al. (1989) found that larger founding populations were more successful for large mammal reintroductions; however, this was true only up to 20–40 founders, after which there was little increase in success rate. Fischer and Lindenmayer (2000) found that reintroductions with more than 100 animals as a founding population in many species resulted in a greater chance of success. Kentucky's high rate of increase, with a current population of 10,000 elk, appears to be related to the high total number of elk released (Table 1); however, when Kentucky is removed from the analysis, there is no relationship between the number of elk released and the rate of increase for the remaining elk populations ($R^2 = 0.03$ without Kentucky; $R^2 = 0.10$ with Kentucky).

Irruptive growth is common in ungulates when the environment is not at carrying capacity and predators and diseases are absent (Riney 1964; Gogan & Barrett 1987). Larkin et al. (2003) suggested that Kentucky's irruptive population growth may eventually be followed by population decline. Irruptive growth has also been seen in other elk populations in Manitoba and Washington State (Banfield 1949; McCorquodale et al. 1988); however, the Washington elk population subsequently

declined and had reduced cow and calf survival (Eberhardt et al. 1996).

Small founder populations may affect more than population growth rates, as they often result in reduced genetic variability due to the founder effect and genetic drift (Conard et al. 2010). However, genetic variability in elk reintroduced across North America was not well explained by founding population size (Conard et al. 2010). Allee (1938) and Armstrong and Seddon (2008) suggested that genetic variability could decrease over time in small populations and initial population growth rates could be negative, as individuals may be too dispersed to find mates (the Allee effect). The distribution of potential mates may therefore be crucial to population growth. In Ontario, 50% of released elk dispersed more than 40 km from their release sites (Yott et al. 2011) and it has been suggested that reducing post-release dispersal of females would enhance calving rates (Larkin et al. 2002). Post-release dispersal can be reduced by holding elk for longer periods of time prior to release to acclimate them to release sites. This has been shown to result in greater post-release site fidelity in Ontario elk (Ryckman et al. 2010).

Table 2. Percent mortality by cause during release years and/or within the first 3 years after the final release (pre-acclimation phase).

Region	N	Hunting/ Poaching	Meningeal Worm	Vehicle/ Train	Predation	Other	Unknown	Source (data from)
Kentucky	410	38.7	45.3	16.0	—	—	—	Dan Crank (2013) (2003–2005)
Bancroft, ON	43	20.9	—	14.0	2.3	62.8	—	Rosatte et al. 2007 (2000–2001)
LH-NS, ON	12	25.0	—	8.3	—	66.7	—	Rosatte et al. 2007 (2001)
Tennessee	62	17.7	8.1	12.9	—	19.4	41.9	Kindall et al. 2011 (2000–2005)
Arkansas ^a	20	35.0	10.0	5.0	—	35.0	15.0	Wes Wright (2013) (1985–1989)
Burwash, ON	101	3.0	—	4.0	40.6	52.4	—	Rosatte et al. 2007 (1998–2001)
LOW, ON	34	29.4	—	2.9	17.6	50.0	—	Rosatte et al. 2007 (2000–2001)

LH-NS, Lake Huron-North Shore.

Other: disease, emaciation, injury, drowning, accidents, and 'other.'

^aMortality within the first 4 years post-release.

Pre-Acclimation Mortality

Prior to release, stress-induced mortality of introduced elk can be substantial. Rosatte et al. (2007) reported that 9% of the mortality in elk reintroduced into Ontario occurred in holding pens prior to release and most of these were related to transport injury and stress. Hunting and poaching played a large role with respect to pre-acclimation mortality, especially in the U.S. populations; however, disease, emaciation, injury, and accidents (classified as other), were the greatest causes of mortality in most cases (Table 2). Predation was more important in the northern populations. The adverse impact of stress is an important component to consider during reintroduction, as many attempts have failed due to stress-related mortality soon after release (Hamr 2001; Teixeira et al. 2007).

Meningeal or brain worm is a nematode that utilizes white-tailed deer (*Odocoileus virginianus*) as the reservoir host and causes no ill effects to this species. However, the worm causes lethal neurological disease in elk, moose (*Alces alces*), and caribou (*Rangifer tarandus*) (Anderson et al. 1966; Anderson 1972; Samuel et al. 1992). Susceptible ungulates become infected by accidentally ingesting gastropods with infective larvae (Anderson 1972). Mortality caused by meningeal worm (*Parelaphostrongylus tenuis*) accounted for 45.3% of reported deaths in Kentucky, 8.1% in Tennessee, and 10% in Arkansas during the pre-acclimation period (Table 2). It has been suggested that past elk reintroduction efforts have failed because of meningeal worm (Raskevitz et al. 1991; O'Gara & Dundas 2002), and it has been predicted that future elk restorations would fail due to mortality caused by this nematode (Carpenter et al. 1973; Severinghaus & Darrow 1976; Bergerud & Mercer 1989; Raskevitz et al. 1991).

As meningeal worms rely on gastropods as their intermediate host, their prevalence may depend on habitats where gastropods are abundant, such as low-lying, damp forests (Anderson 1972). Much of eastern North America has a high density of white-tailed deer and when combined with wetter habitats, gastropods may thrive, thus increasing the prevalence of the nematode (Van Deelen et al. 1997). However, Raskevitz et al. (1991) reported that elk were most often found in habitats with the least number of gastropods. McIntosh et al. (2007) found that within the first few years after reintroduction in south-central Ontario,

59% of deceased elk were infected with *P. tenuis*. Larkin et al. (2003) suggested that because of the high rate of meningeal worm related deaths, the Kentucky elk population would likely decline; however, this has not happened thus far. Bender et al. (2005) showed that elk populations can persist at high levels of productivity in the presence of meningeal worms. This may be because elk are known to survive low levels of meningeal worm infection (Larkin et al. 2003) and do not develop clinical signs in this condition (Samuel et al. 1992; McIntosh et al. 2007). The rate of infection in elk populations may be related to many factors associated with the elk range, such as the parasite prevalence in deer, the abundance of deer, the deer-elk range overlap, the age of infected elk, acquired immunity, and the ability to survive low level infections (McIntosh et al. 2007). Although meningeal worm has accounted for mortality in some of recent elk reintroductions, the negative impact appears lower than initially suspected, indicating that further research of the meningeal worm–elk dynamics would be beneficial.

Vehicle and train collisions accounted for 2.9–16% of post-release elk mortality in eastern North America (Table 2). Although this rate accounts for less mortality than other factors, it deserves attention. Vehicle and train collisions result in alarming numbers of animal deaths every year (Jaren et al. 1991; Romin & Bissonette 1996; Bertwistle 2001; Andreassen et al. 2005). Between 1951 and 1999, 3,791 large animals were killed by vehicle and train collisions in Jasper National Park, of which elk and bighorn sheep made up 53% (Bertwistle 2001). Under these conditions, small populations have greater chances of getting even smaller in accordance with the extinction vortex theory (Gilpin & Soulé 1986). Therefore, in small populations such as reintroduced elk herds where the risk of extirpation is high, every death deserves attention. Developing vehicle and train collision mitigation strategies is likely an important component of ensuring elk restoration success.

Black bears, wolves, coyotes, and cougars are well known elk predators (Singer et al. 1997; Anderson et al. 2005; Barber-Meyer et al. 2008). While wolves tend to focus mainly on adult ungulates for prey (Arjo et al. 2002), black bears and coyotes primarily focus on calves (Carter 2006; Barber-Meyer et al. 2008; Murrow et al. 2009). Black bear predation was the leading cause of elk calf mortality in Great Smoky Mountains

National Park after reintroduction (Murrow et al. 2009). In an environment containing elk, moose, and white-tailed deer, wolves in Riding Mountain National Park in Manitoba preferred elk over other prey, in proportion to their availability (Carbyn 1983). Predator species composition and abundance varied among release regions and predation was found to be a limiting factor in the Burwash and LOW populations of Ontario, accounting for 40.6 and 17.6% of the initial mortality, respectively. These populations have low or negative growth rates, suggesting that predation may be a strong factor influencing population growth. Frair et al. (2007) found that although wolves had an important negative effect on western elk survival during the first post-release winter, elk subsequently learned to mediate their mortality risk, regardless of previous predator experience.

Post-Acclimation Mortality

Major sources of post-acclimation mortality varied among populations (Table 3). Meningeal worm did not account for high mortality, with the exception of the Arkansas and North Carolina populations, which showed positive growth, regardless. From 1981 to 1994, 67–80 years post-release, meningeal worm caused only a small proportion of elk deaths (3% of all mortality) in Michigan (Bender et al. 2005).

Vehicle collisions are currently considered a source of high mortality in some regions. As populations increase, it is likely that more animals come into contact with roads and railways. Even for relatively small populations, collision mortality can have a large negative impact on population dynamics. In recent years, vehicle-wildlife collision mitigation strategies have been put in place for reintroduced elk populations in some areas. For example, in Wisconsin signs along roads that light up when elk are in the vicinity (based on response to radio-collars) have been used. Wisconsin also cut back roadside vegetation to try to allow for early detection of animals close to the road. In the Burwash, Ontario, fencing along a major highway was put up in conjunction with a highway wildlife overpass and highway underpasses, the first of their kind in eastern Canada.

Predation appears to be associated mainly with reintroduced elk populations that have lower rates of increase (Table 2). However, some populations seem to maintain positive rates of increase in spite of predator presence. For example, it has been suggested that in Pennsylvania, elk calves have 80% chance of surviving to 1 year, possibly due to high quality habitat that ensures elk are in top physical condition (DeVivo et al. 2011). McClafferty and Parkurst (2001) state that elk restoration must be based on specific information about the species' historic range, habitat requirements, interspecific relations, socioeconomics, public relations and management, in order to be successful. Conard et al. (2010) suggests that maximizing post-restoration population size, reducing annual variability, and maintaining positive growth rates should assist in promoting the retention of genetic variability. Calf recruitment, an important driver of large herbivore population dynamics

Table 3. Recent major causes of elk mortality in eastern North American reintroductions based on manager surveys.

Region	Hunting/ Poaching	Meningeal Worm	Vehicle/ Train Collision	Predation	Accidental/ Other
Kentucky	*				
Bancroft, ON	*		*		*
LH-NS, ON	*		*		*
Wisconsin			*	*	
North Carolina		*	*	*	
Tennessee ^a	*				*
Arkansas	*	*			*
Michigan	*		*		
Pennsylvania	*		*		*
Burwash, Ontario			*	*	*
Minnesota	*			*	*
LOW, ON				*	*

LH-NS, Lake Huron-North Shore.

^aFrom 2000 to 2005 (Kindall et al. 2011).

(Pimlott 1967; Raithel et al. 2007), is likely influenced by many variables including habitat quality, elk density, predation, and adult male age structure (Gratson & Zager 1998). Larkin et al. (2004) suggested that reintroduction efforts should be focussed on habitats with high levels of open forest edge and limited human disturbance. Areas dominated by a single cover type should be avoided, as they will likely result in lower reintroduction success. Differences among major habitat types in each release location are apparent. For example, Wisconsin's elk range is comprised mainly of unbroken forest (Anderson et al. 2005), while Burwash, Ontario (Popp et al. 2013) and Michigan (Bender et al. 2002) elk ranges consist primarily of forest interspersed with agricultural land and openings.

Elk reintroductions are often poorly documented (Fischer & Lindenmayer 2000), and many programs are plagued with low success and researchers have failed to deliver a scientific framework to managers, which would assist in improving reintroduction success (Deredec & Courchamp 2007). A thorough investigation of factors influencing elk population dynamics in different parts of the continent is needed in order to obtain a clearer understanding of their cumulative effects and to identify those most strongly associated with the success and/or failure of reintroductions (Popp in preparation).

Elk reintroductions have had varying degrees of success in eastern North America; however, more positive results have been displayed during recent years. As can be seen from the varying rates of population growth among recent reintroduction populations, research is still needed to improve on success. In order to address the current lack of knowledge and obtain a better understanding of elk reintroduction dynamics, managers should collaborate and exchange information, post-release monitoring should be intensified, failures should be reported and analyzed, and associated research should continue. Better knowledge should further contribute to the success of elk restoration across the species' former range in eastern North America.

Implications for Practice

- Although the number of successful elk restoration attempts has increased over the last century, there is still a need for the enhancement of reintroduction strategies to ensure that future attempts succeed.
- Major factors contributing to elk mortality during the pre-acclimation and post-acclimation phases of restoration should be considered by managers and mitigation strategies should be developed where necessary.
- Meningeal worm may not be as significant a factor of elk mortality as previously thought; however, further research is needed.

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